

First Presentation Questions

Optimization: what and why?

1. What is optimization? Give an example.
2. What are the three main components of an optimization problem?
3. Explain the difference between:
 - Decision variables
 - Objective function
 - Constraints
4. What does it mean to minimize versus maximize?
5. What is a feasible solution?
6. What is an infeasible solution?
7. What makes a solution optimal?

Mathematical Formulation

8. How do we mathematically formulate an optimization problem?
9. What is the difference between Linear and nonlinear objective functions?
10. What is a feasible region?
11. What is a solution space?
12. Determine the solution space and feasible solution space of the following problem:

$$\max 5x_1 + 7x_2 + 2x_3$$

subject to

$$2x_1 + 4x_2 + 1x_3 \leq 4, \quad x_1, x_2, x_3 \in \{0, 1\}.$$

Continuous vs. Discrete Optimization

13. What is the difference between continuous and discrete optimization?
14. What makes combinatorial optimization different from calculus-based optimization?
15. Give examples of real-life combinatorial optimization problems.
16. Write a linear programming formulation for the following problems:
 - Multi-Dimensional 0/1 Knapsack Problem
 - Minimum Set covering Problem
 - Maximum Independent Set Problem